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Electrodynamic properties of dense semiclassical plasma by molecular dynamics simulation

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In this work a dense semiclassical plasma which occurs in many astrophysical objects and inertial fusion experiments has been investigated. As the result of MD simulation the microscopic and electrodynamic properties of the dense plasma on the basis of effective potentials are presented

1. Interparticle interaction potential.

As it is known, interaction of a couple of charged particles in plasma is different from purely Coulomb interaction. In this case we have some problems with the divergence of collision integrals at big and small scattering angles. The divergence at small angles (long distances) appears as a result of not taking into account the plasma's screening effects on interparticle scattering in the system. Therefore, the method of eliminating divergence at long ranges and correctly describing the properties of strongly coupled plasmas is to use the effective potentials of the interaction of charged particles. In the simple case based on linear response theory we have the well known Debye-Hukkel potential.

The problem of the singularity of Coulomb interaction at the zero distance can be eliminated by introducing quantum mechanical diffraction and symmetry effects at small distances [1-6]. These effects are important when average distance between particles a comparable with thermal wave length of the particles.

The analytical formula in the case of a weakly coupled plasma which takes into account both diffraction and screening effects were obtained in the papers [7,8]. These effective potentials that take into consideration both collective effects at large distances and quantum effects at small ones were obtained using the method of dielectric function. The idea of the method is as follows: we take a microscopic potential that takes into consideration quantum properties but does not consider screening and the Fourier-image of dielectric function is determined based on that microscopic potential. Then, one determines the Fourier-image of effective potential as the ratio of Fourier-images of the micropotential and the dielectric function. Finally, the effective potential is reconstructed with reverse transformations.

In this work micropotential which takes into account the diffraction effect in fully ionized plasma in wide range of the temperature and density are used [9]. For dense plasma this micropotential has the following form:

$$u_{ab}(r) = \frac{e_a e_b}{r} \left\{ 1 - \text{th} \left(\frac{\lambda_{ab}^2}{a_0^2 + br^2} \right) \exp \left[-\text{th} \left(\frac{\lambda_{ab}^2}{a_0^2 + br^2} \right) \right] \right\} \times \left(1 - e^{-r/\lambda_{ab}} \right) - \delta_{ae} \delta_{be} k_B T \ln \left(1 - \frac{1}{2} \exp \left(-\frac{r^2}{\lambda_{ee}^2} \right) \right) \quad (1)$$

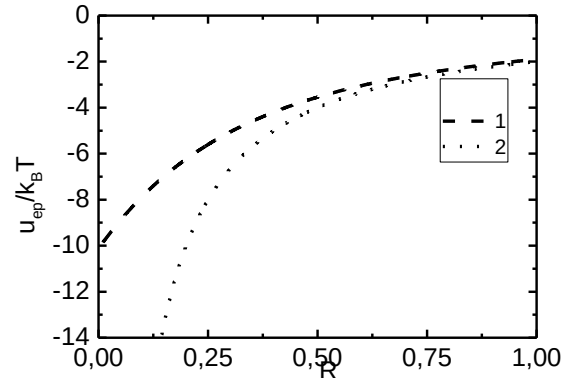


Fig. 6 Proton-electron pair interaction potential (1). Curve 1 is the Deutsch potential[1], 2 is the Coulomb potential. Where $R = r / a_0$, $\Gamma = 2$, $r_s = 9$.

2. Electrodynamic properties of dense plasma.

The fluctuation in dense non ideal plasma can be analyzed through the microscopic excitation spectra. The relation between the wave length and the wave frequency of the collective excitations, can be obtained from the analysis of the autocorrelation (ACF) of the velocity, density and current fluctuations.

Generalized susceptibility can be calculated with help of Fourier transform of the velocity ACF from following formula:

$$\alpha(\omega) = \frac{e^2 n_e}{3k_B T} \int_0^\infty K(t) \exp(i\omega t) dt. \quad (2)$$

The dynamical structure factor ($\vec{k} = k\vec{e}_z$):

$$S(k, \omega) = \frac{1}{2\pi N} \lim_{T \rightarrow \infty} \frac{1}{T} |F_t \{\rho(k, t)\}|^2, \quad (3)$$

where F_t denotes the temporal Fourier transform [10] and the Fourier components of the density fluctuations:

$$\rho(\vec{k}, t) = \sum_{j=1}^N \exp(i\vec{k}\vec{r}_j). \quad (4)$$

The Longitudinal $L(k, \omega)$ and transverse $T(k, \omega)$ fluctuation spectra can be obtained from current operator [11]:

$$\vec{j}(\vec{k}, t) = \sum_{j=1}^N \vec{v}_j(t) \exp(i\vec{k}\vec{r}_j(t)), \quad (5)$$

as follows:

$$L(k, \omega) = \frac{1}{2\pi N} \lim_{T \rightarrow \infty} \frac{1}{T} |F_t \{j_L(k, t)\}|^2, \quad (6)$$

$$T(k, \omega) = \frac{1}{2\pi N} \lim_{T \rightarrow \infty} \frac{1}{T} |F_t \{j_T(k, t)\}|^2. \quad (7)$$

The relation between dynamic structure factor $S(k, \omega)$ and $L(k, \omega)$ [12]:

$$S(k, \omega) = \frac{k^2}{\omega^2} L(k, \omega). \quad (8)$$

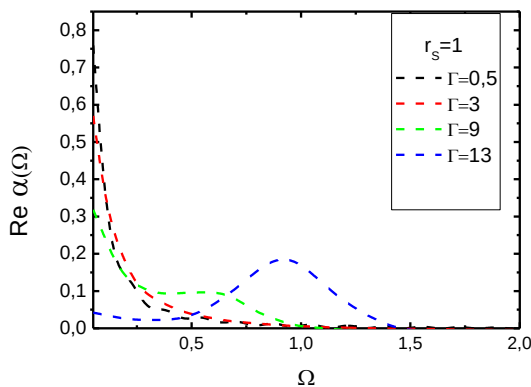


Fig.2 Real part of the generalized susceptibility.

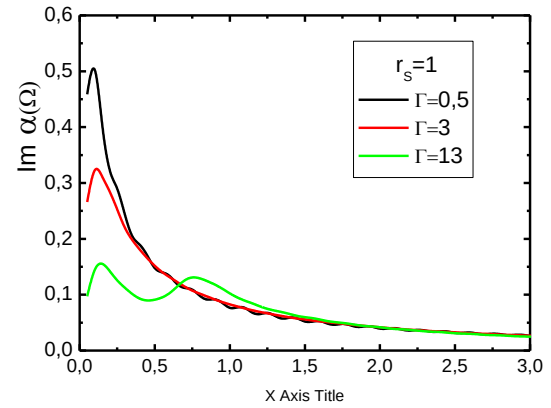


Fig.3 Imaginary part of the generalized susceptibility.

Dynamic dielectric function is calculated from the following formula:

$$\varepsilon(\omega) = \left[1 - i \frac{4\pi\alpha(\omega)}{\omega} \right]^{-1}. \quad (9)$$

On the basis of MD data electrodynamic properties of dense plasma (dynamic structure and dielectric functions, absorption and reflection coefficients, fluctuation characteristics, etc.) have been calculated. The comparison with the results of other approaches is provided.

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